

Single level Inheritance → derived class inherits from one base class

ex- #include <iostream>

Class Animal {

public:

void speak() {

std::cout << "some sound" << std::endl;

}

};

Class Dog : public Animal {

public:

void speak() {

std::cout << "Bark" << std::endl;

}

};

int main() {

Dog d;

d.speak();

return 0;

}

Multiple Inheritance

Ex- #include <iostream>

Class Flyer {

public:

void Fly () {

std::cout << "Flying" << std::endl;

}

};

Class Swimmer {

public:

void swim () {

std::cout << "Swimming" << std::endl;

}

};

Class Dink: public Flyer, public Swimmer {

int main () {

Dink d;

d.Fly();

d.Fly();

return 0;

}

Multilevel Inheritance → a class is derived from another derived class.

ex- #include <iostream>

class Animal {

public:

void speak() {

std::cout << "some sound" << std::endl;

}

};

class Dog: public Animal {

public:

void speak() {

std::cout << "Bark" << std::endl;

}

};

class Puppy: public Dog {

public:

void speak() {

std::cout << "whine" << std::endl;

}

};

int main() {

Puppy p;

p.speak();

return 0;

}

Hierarchical Inheritance → multiple classes are derived from a single base class

ex- #include <iostream>

Class Animal {

public:

void speak() {

std::cout << "some sound" << std::endl;

}

};

class Dog : public Animal {

public:

void speak() {

std::cout << "Bark" << std::endl;

}

};

class Cat : public Animal {

public:

void speak() {

std::cout << "Meow" << std::endl;

}

};

int main() {

Dog d;

Cat c;

d.speak();

c.speak();

return 0;

}

Hybrid Inheritance → combination of two or more types of inheritance

```
#include <iostream>
```

```
Class Animal {
```

```
public:
```

```
void speak() {
```

```
std::cout << "some sound" << std::endl;
```

```
}
```

```
};
```

```
Class Mammal : public Animal {
```

```
public:
```

```
void walk() {
```

```
std::cout << "walking" << std::endl;
```

```
}
```

```
};
```

```
Class Bird : public Animal {
```

```
public:
```

```
void fly() {
```

```
std::cout << "flying" << std::endl;
```

```
}
```

```
};
```

```
Class Bat : public Mammal, public Bird {
```

```
int main() {
```

```
Bat b;
```

```
b.walk();
```

```
b.fly();
```

```
return 0;
```

```
}
```